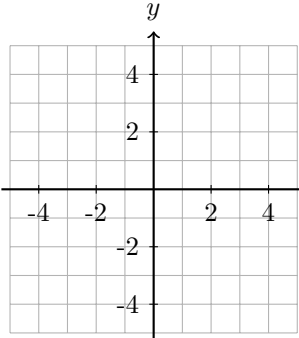


Algebra 2 *Advanced* - Unit 6 Bellwork: Week 3

Name: _____ Date: _____ Period: _____

Day 10: Review - Mixed Graphing + Prior Knowledge

1. Graph $f(x) = -2\sqrt[3]{x+8} + 1$  Transformations: Center point: Zero at $x =$	2. If $f(x) = 2x^3 - 5$, find $f^{-1}(f(3))$ without finding the inverse. Explain your reasoning. Answer: Explanation:
3. If $f(x) = ax + b$ and $f^{-1}(x) = \frac{x-4}{3}$, find a and b. $a =$ $b =$	4. [SAT] A system has infinitely many solutions. One equation is $3x - 2y = 12$. Which could be the other? A) $6x - 4y = 12$ B) $-6x + 4y = -24$ C) $3x + 2y = 12$ D) $6x - 4y = 18$ Answer:

Day 11: Review - Finding Inverse Functions

4 Steps: (1) Replace $f(x)$ with y (2) Swap x and y (3) Solve for y (4) Write as $f^{-1}(x)$

1. Find $f^{-1}(x)$ for $f(x) = \frac{2x-3}{5}$ $f^{-1}(x) =$	2. Find $f^{-1}(x)$ for $f(x) = (2x-1)^3 + 4$ $f^{-1}(x) =$
3. Find $f^{-1}(x)$ for $f(x) = 3\sqrt[3]{x+2} - 6$ $f^{-1}(x) =$	4. [SAT] $g(x) = f(2x) + 3$ where $f(x) = x^2 - 1$. What is $g(-2)$? A) 2 B) 6 C) 12 D) 18 Answer:

Day 12: Review - Verifying Inverses

$$f(g(x)) =$$

$$g(f(x)) =$$

Inverses?

2. Find $f^{-1}(x)$ for $f(x) = \frac{x^3 + 1}{8}$

$$f^{-1}(x) =$$

Hint: What is the inverse of the inverse?

$$a =$$
$$b =$$

A) 5 B) 10 C) $10 - k$ D) Cannot determine

Answer: